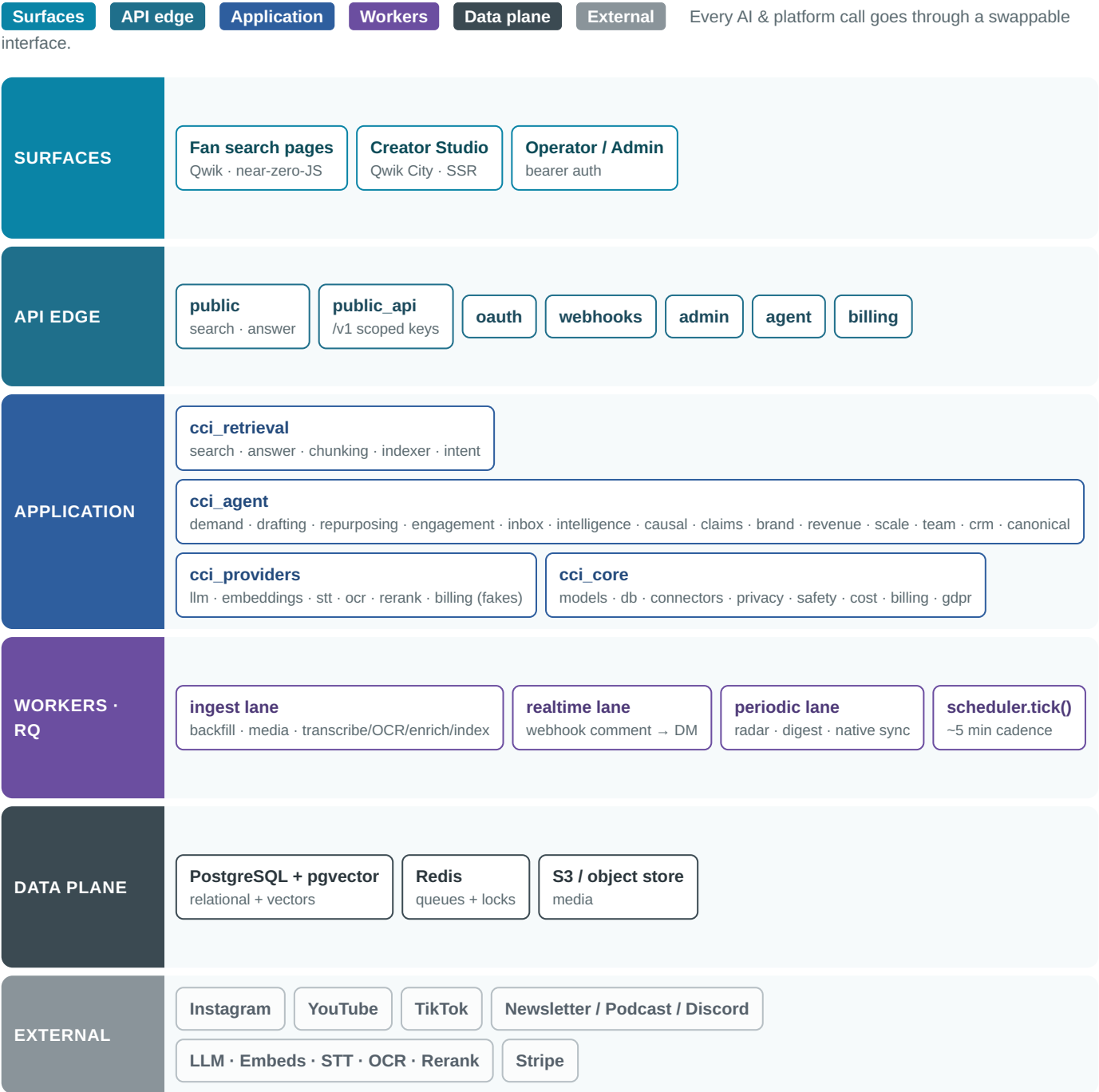


Sift — Architecture & Workflow Pipeline

A cross-platform demand-to-outcome platform for creators · system overview and end-to-end data flow

1 · System architecture (layered)



3 · The complete workflow pipeline (end to end)

From a creator connecting an account to a closed, measured demand loop. Tags show the lane / surface each stage runs on.

- 1 Connect** oauth · admin
Creator OAuth or operator token; capability-tagged account (incl. TikTok per-account full-loop).
`oauth.py · admin.set_token · PlatformAccount · connectors registry`
- 2 Sync / Ingest** ingest lane
Manual 'Sync now', 'Sync all', or hourly. Connector output → Posts + Comments, idempotent, per-item SAVEPOINT, PII-redacted + pseudonymized.
`connectors.backfill_content → sync_native / ingest_instagram / ingest_youtube / ingest_import`
- 3 Enrich** ingest lane
Per new post: transcribe (Whisper), OCR on-screen text, LLM summary/topics/keywords, language detection.
`enrich.py · providers (stt, ocr, llm) · language.py → Transcript`
- 4 Index** ingest lane
Chunk content + embed into pgvector beside a generated tsv full-text column. Deletion purges chunks.
`chunking.py · indexer.py · embeddings provider → Chunk(embedding, tsv)`
- 5 Serve** API · public
Hybrid search: FTS + pgvector cosine → RRF fusion → optional rerank → confidence. Grounded cited answer card, or 'no strong answer'.
`search.py (RRF + rerank) · answer.py (cited; voice = post-retrieval rewrite)`
- 6 Listen** realtime lane
Comment webhook (signature-verified, idempotent) → intent detection → same retrieval layer. Query logged as demand.
`webhooks.py · intent.py · _claim_event (dedup)`
- 7 Respond** realtime lane
Approval-mode DmJob → dispatcher: capability gate + per-creator advisory lock + 7-day window + hourly cap. One public reply + one private DM (answer + deep link); low-confidence → archive link.
`dm.dispatch_approved · _can_dm · _try_lock · deep_links.py`
- 8 Learn** periodic lane
Cluster searches + comments + external signals → Demand Integrity (unique askers, organic/prompted, persistence, sentiment, manipulation risk, exposure-normalized) → explainable Opportunity Score.
`radar.build_radar · demand.compute_integrity · opportunity_score → DemandTopic`
- 9 Reconcile** API · admin
Each card classified against the creator's own archive: gap / partial / well, with the covering permalink — 'make new' vs 're-promote'.
`_reconcile_demand · coverage strength + permalinks`
- 10 Act** agent layer
One-tap: content brief + draft (script/hooks), series builder, multi-format repurpose + Sift & Shift, reply assistant, Loop-Closer, briefing.
`drafting · repurposing · engagement · briefing · canonical`
- 11 Close** state machine
Publish or re-promote → demand state → loop_closed; Outcome logged. North-star: closed loops/week, split made vs re-promoted.
`agent_foundations.transition_demand · repromote_topic · Outcome`
- 12 Measure** eval · cost
Citation-correctness gate (fails closed if nothing answerable) before automation; events + per-creator AI cost metering; causal layer (holdouts, baselines, CIs) measures lift.
`cci_eval.harness · events.py · cost.meter_llm · causal.py`

4 · Key sub-pipelines

A · Ingestion — content becomes searchable



B · Retrieval & answer — one hybrid layer serves search, the answer card, and DM intent



C · Demand → action loop — the monetizable asset



5 · Jobs, queues & scheduler

Lane	What runs	Cadence
ingest	Backfills, media download, transcribe / OCR / enrich / index	On connect & on demand
realtime	Webhook comment → intent → DM job; approved-DM dispatch (advisory-locked)	Webhook + every tick
periodic	Demand Radar build, weekly digest, incremental + native sync, token refresh	Scheduled

scheduler.tick() (~5 min): DM dispatch every tick · hourly incremental + native sync · weekly radar + digest (Mon 07:00 UTC) · daily OAuth token refresh (06:00 UTC).

6 · Cross-cutting concerns

Concern	How it's enforced
Security	Creator-scoped queries; OAuth tokens encrypted at rest (EncryptedString); fail-closed boot on insecure secrets; webhook signature verified (fail-closed in prod); prompt-injection hardening (wrap_untrusted + GUARD) on agent prompts; DM dispatch advisory lock.
Privacy / GDPR	Per-creator pseudonyms (dedicated secret); PII redaction before storage; export + erase (incl. erase-by-pseudonym); opt-in row-level security.
Cost	Per-creator AI metering with op tags; per-plan monthly cost cap; separate anonymous-answer abuse ceiling (free card stays free).
Portability	Every AI + platform call behind an interface (offline fakes by default); Postgres + pgvector; S3; Alembic migrations 0001–0004; k8s + Render manifests.
Quality	Citation-correctness eval gate before automation; 311 backend + 9 frontend tests; five-dimension adversarial review.